

Effects of 8 weeks sustained follow-up after a nurse consultation on hypertension: A randomised trial

Caroline Wai Chiu ^a, Frances Kam Yuet Wong ^{b,*}

^a Family Medicine and General Outpatient Clinic, Hong Kong SAR, China

^b School of Nursing, The Hong Kong Polytechnic University, Hong Kong SAR, China

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ABSTRACT

Background: The prevalence of hypertension is high, but the overall control rate is low. Poor control of, hypertension is associated with a number of diseases, such as stroke, heart and renal failure, and high, mortality rates. Studies have shown the separate effects of nurse clinics and telephone follow-up on, blood pressure control, but the incremental effect of combining the two interventions is unknown.

Objectives: This study examines whether there is an incremental effect on blood pressure control when using a nurse clinic combined with telephone follow-up.

Methods: This was a randomised controlled trial. The primary outcome measure was blood pressure reading. The secondary outcome measures included adherence to home blood pressure monitoring, exercise, diet, medication, and satisfaction with care.

Results: There were no significant differences in the baseline measures between the control and study groups. Significant differences were found at 8 weeks after intervention was initiated between groups in, systolic blood pressure (control -7.97 vs study -19.03 , $t = 2.35$, $p = 0.022$, CI 1.66 – 20.47) and diastolic, blood pressure (control -3.72 vs study -11.68 , $t = 3.02$, $p = 0.004$, CI 2.68 – 13.24). Other variables with a significant between-group differences ($p < 0.05$) were blood pressure control rate, adherence, to home blood pressure monitoring, exercise and satisfaction with care. Further analysis using, regression showed that home blood pressure monitoring is the most significant predictor for improved, systolic blood pressure.

Conclusions: This study showed that nurse clinics have positive effects on blood pressure control and adherence to healthy lifestyle, but telephone follow-up after such clinics augments the effects of the clinic consultation. This combined mode of services is worth considering for other chronic disease, management programmes.

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What is already known about the topic?

- Blood pressure control is generally poor
- Nurse clinics, nurse telephone follow-up and adherence to healthy lifestyle are effective measures to help control blood pressure

What this paper adds

- Blood pressure control can be enhanced by a nurse clinic but nurse telephone follow-up after clinic consultation augments the effects for blood pressure control
- A nurse clinic with telephone follow-up is also effective in enhancing a healthy lifestyle, which is important for chronic disease management. Evidence from this study is therefore worth considering for other chronic disease management programmes

* Corresponding author. Tel.: +852 27666419.

E-mail address: hsfwong@inet.polyu.edu.hk (F.K.Y. Wong).

1. Introduction

The prevalence of hypertension is high but the overall control rate is low. Poor control of hypertension is a high risk for a number of diseases such as stroke, heart and renal failure. Effective blood pressure (BP) control is important to the health of individuals and cost containment of the healthcare system. Since the BP control rate is maintained at less than 50% on average, more effort is needed to enhance the control effect.

Hypertension is defined as a systolic BP > 140 mmHg, a diastolic BP > 90 mmHg, or using anti-hypertensive medication, or being told twice by a healthcare professional that one has hypertension (Fields et al., 2004; American Heart Association, 2009). The prevalence rate is similar across countries, for instance, 27.2%, 30.0% and 31.3%, respectively, for China (Gu et al., 2002), the United Kingdom (Falaschetti et al., 2009) and the United States (American Heart Association, 2009). However the awareness, treatment and control rates vary. The reported rates respectively for China (Gu et al., 2002), the United Kingdom (Falaschetti et al., 2009) and the United States (American Heart Association, 2009) are 44.7%, 66.0%, 78.7% (awareness), 28.2%, 54.0%, 69.1% (treatment), and 8.1%, 28%, 45.4% (control).

The BP control rates are far from satisfactory. Hypertension is well documented as having strong associations with coronary heart disease, stroke, congestive heart failure, and end-stage renal disease (He et al., 2002; Hodgson and Cai, 2001). Wyatt et al. (2008) reported that a fall in systolic BP of 10–20 mmHg sustained for 5 years could reduce the risks of myocardial infarction by 25% and stroke by 40%. It is estimated that over US \$100 billion has been spent in 1 year on controlling hypertension (Hodgson and Cai, 2001).

There are a number of demographic variables associated with hypertension. Africans and South Asians have a higher prevalence rate compared with Caucasians (Agyemang et al., 2005). Women and older adults are more likely to have hypertension compared to men and younger persons (Fields et al., 2004; Wyatt et al., 2008). While the demographic variables are not mutable, measures involving healthcare professionals to provide regular monitoring and reinforcement of clients' health behaviours were found to be helpful in controlling high BP. Having a regular primary care provider (Victor et al., 2008) and insurance that ensures access to care (Bleich et al., 2007) were found to be significantly related to better BP control. In a review of six randomised controlled trials of nurse-led cardiac clinics, systolic and diastolic BP were significantly reduced in the nurse-led groups as compared to the usual care groups (Page et al., 2005). An economic analysis study showed that nurse clinics were cost-effective and good value for money for lowering BP involving diabetic patients (Mason et al., 2005).

Besides direct consultation with healthcare providers, the practice of home BP measurements was also found to be significantly associated with better BP control (Rogers et al., 2001; Obara et al., 2008). A recent study found that having a nurse who provided subsequent telecounselling about lifestyle modification and medication upon receiving the readings seemed to have a more significant effect in

reducing systolic BP than mere reporting of results (Artinian et al., 2007). A cluster analysis revealed that patients who had higher adherence to medications and healthy lifestyle regimens were more likely to have better health outcomes, including BP control (Weir et al., 2000).

The literature has revealed that regular clinic consultations, home BP monitoring, and home BP monitoring with subsequent telephone follow-up are all effective measures to help control BP. The rate of BP control, however, continues to be low, and more attempts are needed to enhance the control effect. This study was therefore launched to see if the combined efforts of a nurse clinic and telephone follow-up would augment the therapeutic effects of BP control.

2. The study

2.1. Design

This study was a randomised controlled trial. Patients who fit the criteria were randomised to the study or control group using sets of computer-generated random numbers. The numbers were placed in sealed envelopes and the allocation of treatment was only known after the patient entered into the study. Please see Appendix A for the study flow.

2.2. Research questions

In this study, we asked whether the study group receiving telephone follow-up after a nurse clinic consultation was different from the control group who only received nurse clinic consultation. The outcome measures were BP readings, adherence to home BP monitoring, healthy lifestyle and medication, and satisfaction with care.

2.3. Setting and participants

Subjects were recruited from a family medicine clinic in the public hospital system of Hong Kong from August 2008 to February 2009.

Inclusion criteria:

- Referred by physicians at the family physician clinic with poor BP control or newly diagnosed with hypertension
- Confirmed to have a reading of systolic BP over 140 mmHg or diastolic BP over 90 mmHg at the nurse clinic
- Able to communicate
- Alert and oriented
- Able to be contacted by telephone at home

Exclusion criteria:

- Not able to communicate, e.g. not able to speak the local language, or having hearing difficulties
- Old-age home residents

2.4. Ethical considerations

Ethical approval was obtained from the Human Subjects Ethics Committee of the study hospital and

university. Information and explanation were provided to the subjects, and those who agreed to participate in the study were asked to sign a consent form.

2.5. Intervention

The study group received both routine BP nurse clinic consultation and nurse-initiated telephone follow-up after the clinic. The control group received the nurse clinic consultation only. The nurse clinic was run by two experienced nurses who had worked in primary health care settings for more than 8 years. During the study period, one nurse would only provide consultation to the control group and the other only to the study group. However, the nurse clinic had set protocols and both nurses were audited for consistent approaches and compliance with the set protocols prior to conducting the study. The protocols were validated by a primary healthcare expert panel, including two consultants of family medicine, a nurse manager and two experienced nurses in the field. The inter-rater reliability of the client assessments and documentations was confirmed using Kappa statistics ($k = 0.852$, $p = 0.005$). The set of protocols included assessments of knowledge and behaviour in BP control, use of medication, BP home monitoring, exercise, diet, smoking, alcohol consumption, body weight control and stress management. If further interventions such as treatment of co-morbidities or stress management were needed, the clients would be referred to other members in the healthcare team as appropriate. At the end of the nurse clinic consultation, the nurse would set mutual goals with the client and arrange the next nurse clinic appointment for 8 weeks later. The consultation session would usually last about 45 min.

In the study group, the nurse would provide two telephone calls to the clients every 2–3 weeks during the span of 8 weeks after the index nurse clinic and before the next clinic visit. The calls were guided by a nurse-initiated protocol with a structured format as follows: (1) nurse's self-introduction, general addressing of the patient's health condition, (2) assessing patient's BP readings, adherence to healthy lifestyle and reviewing contracted goals, (3) reinforcing health self-management behaviours, providing health advice, and assessing the need for referrals, (4) reviewing and revising, if appropriate, the mutually set health goals, and (5) ending the telephone call using words of encouragement and arranging the time for the next call, except for the last call.

2.6. Sample size

We computed the required sample size based on the study by [Denver and Barnard \(2003\)](#), who showed that the study group with nurse clinic follow-up had a mean systolic BP difference of 10. The calculated sample size was 68 in each group to achieve a power of 0.8 and a significance level of 0.05. The actual study could only recruit a total of 63 subjects (32 control, 31 study) because of the need to re-deploy staff due to an unforeseeable incident of toxic milk products in the community during the study period. This sample size achieved a power of 0.5 based on the study by [Denver and Barnard \(2003\)](#).

2.7. Measures

2.7.1. Study protocol

The study protocol contained basic demographic and health information used for baseline comparison. The primary outcome measure, the BP reading, was measured at the clinic session. BP was measured using a Welch Allyn digital monitor and the equipment was calibrated by the engineering team of the clinic. Measurements were carried out in both arms, and if no significant difference was found, the measurement of the right arm was used. The significant difference was defined as a difference of over 10 mmHg for the systolic pressure. If such a difference occurred, the BP would be re-checked after the client rested for 10 min. When a significant difference was still detected, the client would be referred for medical consultation and excluded from this study.

2.7.2. Adherence assessment form

This form contained information on adherence to home blood pressure monitoring, exercise, diet and medication (for clients on drug treatment). The form was designed referencing [Wong et al.'s \(2005\)](#) work on diabetic control and validated by the same expert panel mentioned in the section above. Panel members were asked to indicate the degree of relevance of the items and the index of content validity was 0.92. The inter-rater reliability between the two nurses who ran the clinics was confirmed using 5 pilot cases ($k = 0.82$, $p = 0.005$). The nurse who ran the clinics assigned a score for each aspect of adherence according to the description below, based on the assessment of clients' self-reported information obtained in the consultation.

2.7.2.1. Home blood pressure monitoring. This measure had two aspects, including the frequency (e.g. every week) and time (e.g. in the morning) of BP monitoring according to the set goals. A score of '1' was given to adherence and '0' to non-adherence for each aspect. The maximum score was 2 and the minimum was 0.

2.7.2.2. Exercise. The hypertension counselling protocol recommended that exercise should be performed for at least 30 min/day, or at least 1 h/alternate day or for an accumulated 30 min/day (not less than 10 min/time). A score of '2' was given to clients who completely met the exercise recommendations, '0' for not doing any exercise and '1' for partial fulfilment of the exercise recommendations.

2.7.2.3. Diet. The diet adherence included aspects of salt restriction, control of fat intake, and adequate fruit and vegetable consumption. A score of '2' was given to clients who completely met the diet recommendations, '0' for fulfilling none of the recommendations and '1' for partial fulfilment of the diet recommendations.

2.7.2.4. Medication. The aspects of medication adherence included dose, frequency, and timing of taking the anti-hypertensive medication if applicable. A score of '1' was given to adherence and '0' to non-adherence for each aspect. The maximum score was 3 and the minimum was 0.

2.7.3. Satisfaction

The satisfaction questionnaire was adapted from the one used in Wong et al. (2005) study and the content of the items was modified to make it relevant to this study. The questionnaire contained 9 items. They were aspects of hypertension education, intervention, follow-up and service from the nurses measured on a 5-point Likert scale, from '5' strongly agree to '1' strongly disagree. The maximum score was 45 and the minimum was 9. Test-retest reliability over 10 days was confirmed among the 5 pilot cases who were also involved in trying out the adherence assessment form mentioned above ($k = 0.84$, $p = 0.005$).

2.8. Data collection

The data were collected by the two nurses at the index nurse clinic visit and the subsequent nurse clinic visit at 8 weeks as part of the clinic assessment data. The satisfaction questionnaire was delivered by a clinic assistant who was blind to the grouping prior to the second nurse clinic visit.

2.9. Data analyses

Baseline data between the study and control groups were compared using the Chi-square and independent t -tests. For comparison of outcome measures within the groups, paired t -tests and the Wilcoxon signed test were used to compare respectively the differences of means and relative ranking of values. For comparison between groups, independent t -tests and the Mann–Whitney U -test were used to compare respectively the differences in means and ranked scores. Multivariate linear regression analysis was executed to detect factors contributed to improved systolic BP. All the data were analysed using SPSS 11.1 for Windows.

3. Results

3.1. Baseline data

There was no significant difference in the baseline characteristics between the study and control group (Table 1).

Table 1
Comparison of the characteristics of the two groups.

Characteristics	Total ($n = 63$)		Control ($n = 32$)		Study ($n = 31$)		Statistical tests	
	n	(%)	n	(%)	n	(%)	Value	p
Gender								
Female	42	(66.7)	18	(56.3)	24	(77.4)	3.175	0.075 ^a
Male	21	(33.3)	14	(43.8)	7	(22.6)		
Education level								
No education	4	(6.3)	4	(12.5)	0	(0)	5.022	0.17 ^a
Primary	28	(44.4)	14	(43.8)	14	(45.2)		
Secondary	27	(42.9)	13	(40.6)	14	(45.2)		
Tertiary or above	4	(6.3)	1	(3.1)	3	(9.7)		
Financial support								
CSSA (Comprehensive Social Security Assistance)	4	(6.3)	3	(9.4)	1	(3.2)	2.516	0.472 ^a
Employed	25	(39.7)	10	(31.2)	15	(48.4)		
Family support	34	(54.0)	19	(59.4)	15	(48.4)		
Family history of hypertension								
Yes	36	(57.1)	14	(43.8)	22	(71)	3.717	0.54 ^a
No	27	(42.9)	18	(56.3)	9	(29)		
Co-morbidity								
Yes	5	(7.9)	4	(12.5)	1	(3.2)	0.802	0.37 ^a
No	58	(92.1)	28	(87.5)	30	(96.8)		
Smoking								
Yes	6	(9.5)	4	(12.5)	2	(6.5)	1.51	0.69 ^a
No	57	(90.5)	28	(87.5)	29	(93.5)		
Alcohol ^b								
Yes	5	(7.9)	3	(9.4)	2	(6.5)	0	1 ^a
No	58	(92.1)	29	(90.6)	29	(93.5)		
On hypertensive medication								
Yes	30	(45.0)	14	(43.8)	16	(51.6)	0.39	0.532 ^a
No	33	(55.0)	18	(56.3)	15	(48.4)		
Age (years), mean (SD)	53.87	(7.65)	54.4	(7.57)	53.3	(7.82)	0.59	0.59
Body weight (kg), mean (SD)	62.14	(11.8)	64.3	(12.3)	59.9	(11)	0.59	1.48
Body mass index, mean (SD)	24.67	(3.5)	25.01	(3.3)	24.32	(3.73)	0.59	0.77

^a Chi-squared test.

^b Data of one subject missing.

^c t -test.

Table 2

Comparison of the blood pressure readings and control between the two groups.

	Control (<i>n</i> = 32)		Study (<i>n</i> = 31)		Statistical tests		95% CI
	Mean	(SD)	Mean	(SD)	<i>p</i>		
SBP (mmHg)							
Pre-test	148.50	(17.93)	147.19	(18.95)	0.28	0.78 ^a	−7.80–10.61
Post-test	140.53	(16.59)	128.16	(15.66)	3.04	0.003 ^a	4.25–20.50
Difference in SBP	−7.97	(20.30)	−19.03	(16.91)	2.35	0.022 ^a	1.66–20.47
<i>t</i> , <i>p</i> -value	2.220	0.034	6.267	<0.0001			
DBP (mmHg)							
Pre-test	88.84	(11.85)	90.48	(11.44)	−0.56	0.58 ^a	−7.51–4.23
Post-test	85.12	(13.67)	78.81	(9.08)	2.15	0.035 ^a	0.48–12.16
Difference in DBP	−3.72	(10.63)	−11.68	(10.31)	3.02	0.004 ^a	2.68–13.24
<i>t</i> , <i>p</i> -value	1.979	0.057	6.306	<0.0001			
BP control under 140/90 mmHg at 8 weeks	<i>n</i> (%) = 24(75.0)		<i>n</i> (%) = 8(25.8)		8.847		<i>p</i> = 0.003 ^b

^a *t*-test.^b Chi-squared test.

3.2. Differences in blood pressure reduction and blood pressure control

The primary outcome for this study was the difference in reduction of BP. There was no statistical difference in the baseline data in terms of systolic and diastolic BP between the study and control groups. The control group had a significant reduction in systolic BP and the study group achieved significant reduction for both systolic and diastolic BP over a period of 8 weeks. The difference from baseline to 8 weeks respectively for systolic and diastolic BP, was –7.97 mmHg (*p* = 0.034) and –3.72 mmHg (*p* = 0.057) for the control group and –19.03 mmHg (*p* < 0.0001) and –11.68 mmHg (*p* < 0.0001) for the

study group. The improvement was much more prominent in the study group when compared with the control group and there was a statistically significant difference (systolic BP *t* = 2.35, *df* = 61, *p* = 0.022, diastolic BP *t* = 3.02, *df* = 61, *p* = 0.004). We also computed the BP control rate for both groups after the intervention period. The rates for the control and intervention groups were respectively 25.8% and 75.0% (χ^2 = 8.847, *p* = 0.03) (Table 2).

3.3. Self-care adherence

The adherence scores were equivalent between the study and control groups at the baseline for home blood pressure monitoring (HBPM), exercise, diet and medica-

Table 3

Comparison of self-care adherence between the two groups.

Characteristics	Control (<i>n</i> = 32)		Study (<i>n</i> = 31)		<i>Z</i>	<i>p</i>
	Median	IQR ^a	Median	IQR ^a		
HBPM ^b						
Pre-test	2	(2–2)	2	(2–2)		
Post-test	1	(1–1)	1	(1–1)		
Difference in HBPM between groups	2	(0–3)	3	(3–3)	–3.94	<0.0001 ^c
<i>Z</i> , <i>p</i> -value	–4.47	<0.0001 ^d	–5.29	<0.0001		
Exercise adherence						
Pre-test	0	(0–2)	0	(0–1)		
Post-test	1.5	(1–2)	2	(2–2)		
Difference in exercise adherence between groups	0	(0–1)	2	(1–2)	–4.51	<0.001 ^c
<i>Z</i> , <i>p</i> -value	–3.63	<0.0001 ^d	–4.60	<0.0001		
Diet adherence						
Pre-test	2	(2–2)	2	(1–2)		
Post-test	2	(2–2)	2	(2–2)		
Difference in diet compliance between groups	0	(0–0)	1	(0–1)	–1	0.32 ^c
<i>Z</i> , <i>p</i> -value	–1.41	0.157 ^d	–2.71	0.007		
Medication adherence						
Pre-test	3	(3–3)	3	(2–3)		
Post-test	3	(3–3)	3	(3–3)		
Difference in drug compliance between groups	0	(0–0)	0	(0–1)	–1.187	<0.235 ^c
<i>Z</i> , <i>p</i> -value	–1.41	0.157 ^d	–2.00	0.046		

^a IQR, interquartile range.^b HBPM, home blood pressure monitoring.^c Mann–Whitney *U*-test.^d Wilcoxon signed rank test.

tion. The control group achieved significantly improved scores in HBPM ($Z = -4.47$, $p < 0.0001$) and exercise ($Z = -3.63$, $p < 0.0001$) over time but not in diet ($Z = -1.41$, $p = 0.157$) and medication ($Z = -1.41$, $p = 0.157$) adherence. The study group made significant improvements over time for HBPM ($Z = -5.29$, $p < 0.0001$), exercise ($Z = -4.60$, $p < 0.0001$) and diet ($Z = -2.71$, $p = 0.007$), and marginal significance for medication adherence ($Z = -2.00$, $p = 0.046$). When comparing between groups, significant statistical differences were achieved for HBPM ($Z = -3.94$, $p < 0.0001$) and exercise ($Z = -4.51$, $p < 0.001$) adherence (Table 3).

3.4. Satisfaction

The study group had a higher satisfaction score (m 39.32, SD 5.30) than the control group (m 29.59, SD 7.14), and there was a significant difference ($t = 6.17$, $df = 61$, $p < 0.001$).

3.5. Predictor for improved systolic BP

We executed multivariate linear regression analyses to find out what factors contributed to improved systolic BP. Variables for which p -value was < 0.1 in the univariate analyses were put in the regression model for stepwise selection. Eventually, there was only one factor that came out to be significant in the final model in predicting improved SBP, and that was HBPM adherence (beta = -0.34 , $p = 0.006$). The explanatory power of this model was however weak ($R^2 = 0.115$).

4. Limitations of the study

We acknowledge that there are at least four limitations to this study. The subjects included in the study were in the early stage of being diagnosed with hypertension, and the effect of the intervention adopted in this study needs to be further tested in patients who have a longer history of hypertension and possibly more co-morbidities. Also, the sustained effect of the intervention needs to be established in further studies. This study was confined to one family clinic with a small sample size and a relatively weak power of 0.5. The generalisation of the results has its limitations. In addition, we did not have a mechanism to validate whether the self-reporting of adherence to a healthy lifestyle truly reflected the actual practice. This study also suffered from the fourth limitation, that is, the use of two different nurses to provide care for the control and study group. Though we tested the inter-rater reliability of client assessments and documentations of the two nurses, we could not be sure of the individual differences that might occur in care.

5. Discussion

This study provided evidence to support that although nurse clinics are effective in enhancing BP control, the effects are augmented with subsequent telephone follow-up. In addition to BP control, the telephone follow-up after the visit to the nurse clinic helped to enhance adherence to

a healthy lifestyle, including HBPM and exercise. The satisfaction with care was also significantly higher in the study group. It is interesting to detect in the regression model that HBPM is the only factor that predicts BP control. In other words, empowering clients to monitor their health condition is very important for health maintenance. Effective BP control can help saving costs and promoting individual well-being as the literature consistently reports an unsatisfactory BP control rate as low as 8.1% and on average no more than 50% (Gu et al., 2002; Falaschetti et al., 2009; American Heart Association, 2009). Effective BP control rate remains a challenge for healthcare providers. The results of this study suggest that chronic care management is a continuous effort that requires sustained follow-up and involvement of clients in self-monitoring.

Successful control of hypertension would not only help reduce health costs but also enhance individual well-being. A study by Wang et al. (2007) has shown that drug treatment in itself could not explain better BP control. In the study treatment differences in Western Europe and the United States were compared and the United States was found to have better hypertension control. Patients in the United States started treatment at a lower threshold and was more intensive. BP control, however, was not found to be associated with cross-national patterns of anti-hypertensive drug selection and consequently drug cost differences. In other words, there was no evidence supporting the relationship between drug treatment and BP control. Wang et al. (2007) suggest that BP monitoring and regular follow-up care are more important in hypertension control. Findings in this study support this hypothesis. Intervention employed in this study involves sustained follow-up, focussing on lifestyle modification and reinforcement of appropriate health behaviours. Our study reveals a reduction of 19.0 mmHg in systolic BP over 8 weeks in the intervention group. Similar results were found among a group of general hypertensive patients, where they were adherent to medication and lifestyle regimens had better health outcomes (Weir et al., 2000). Using data from a US national survey, type 2 diabetes patients who were highly adherent to medication treatment and lifestyle modification regimens were also more likely to achieve BP control (Suh et al., 2009). This study has isolated a significant variable that predicts the improvement of SBP, that is, adherence to HBPM. This is substantiated by the study by Obara et al. (2008) who showed that self-measurement of BP at home is significantly associated with better BP control. If the clients maintain their BP control over 4–5 years, that helps to reduce their risks of coronary heart disease and stroke up to 25% and 40%, respectively (Artinian et al., 2007; Wyatt et al., 2008).

The achievement of a high BP control rate in our study (75.0%) is consistent with findings in a review (Page et al., 2005) of 6 randomised controlled trials where a rate of 72.9% was reported. The BP control rate of 25.8% in the control group in our study is similar to the BP control reported from the United Kingdom (Falaschetti et al., 2009).

Evidence obtained from this study demonstrates that nurse clinics and nurse follow-up are effective care

measures for clients with hypertension. Studies involving other disease groups, such as cardiac (Page et al., 2005), diabetes (Wong et al., 2005), renal (Wong et al., 2009), cancer (Loftus and Weston, 2001), and continence (Williams et al., 2000), also prove that nurse clinics and nurse follow-up are important health service strategies for chronic disease management. In nurse-led services, the key interventions applied are health assessment, health teaching/counselling, and making referrals as appropriate (Page et al., 2005; Wong and Chung, 2006), and the outcomes of interest are management of symptoms, prevention of complications, and client satisfaction (Wong and Chung, 2006). The nurse adopts a holistic approach in care, which addresses the health needs of the clients in relation to their individual lifestyle and social context (Page et al., 2005; Wong and Chung, 2006). This approach is an interactive process involving the clients as partners and coaching them to take ownership in managing their own health conditions. This patient empowerment approach is advocated in chronic care models (Wagner, 2000; Schillinger et al., 2009). This study suggests that the effects of a nurse clinic can be augmented by supplementing with nurse-initiated calls to identify the existence of symptoms, the degree of adherence to healthy behaviours, and to provide management accordingly, such as reinforcement of appropriate behaviours, health counselling and making referrals. The measure of sustained follow-up is included as one of the key measures for effective chronic illness programmes by Wagner (2000), and telephonic nursing is an effective mode of providing sustaining services for health maintenance. Telephonic nursing has the advantage of promoting efficient accessibility, that is, being able to reach a relatively large number of clients at a low cost, particularly for clients who have mobility problems and live far away from care facilities (Lorentz, 2008). A study has compared three discharge care models using home telecare with a video-conferencing device, nurse telephone calls and usual outpatient care (Jerant et al., 2001). Both intervention groups had significantly better outcomes in terms of readmissions and associated charges when compared with the usual care group, but the

home telecare group did not offer incremental benefit beyond telephone follow-up. In other words, nurse telephone calls are as effective as home telecare, but the cost is much lower (Jerant et al., 2001).

6. Conclusion

Previous studies have provided evidence to show the effectiveness of nurse clinics and nurse telephone follow-up, but there is a lack of studies assessing the incremental benefits of telephone follow-up after nurse consultations. The current study has demonstrated that nurse clinics are effective, but telephone follow-up further augments the effects. Control of BP is essential to delay clients' progression to cardiovascular problems and stroke events, and even death. Effective BP control using non-pharmacological treatment modes can also reduce the costs of medication use such as side-effects associated with the drugs. This mode of care delivery, using nurse telephone follow-up to supplement nurse clinic consultations, is worth considering for other chronic disease management programmes.

Conflict of interest

None.

Funding

None.

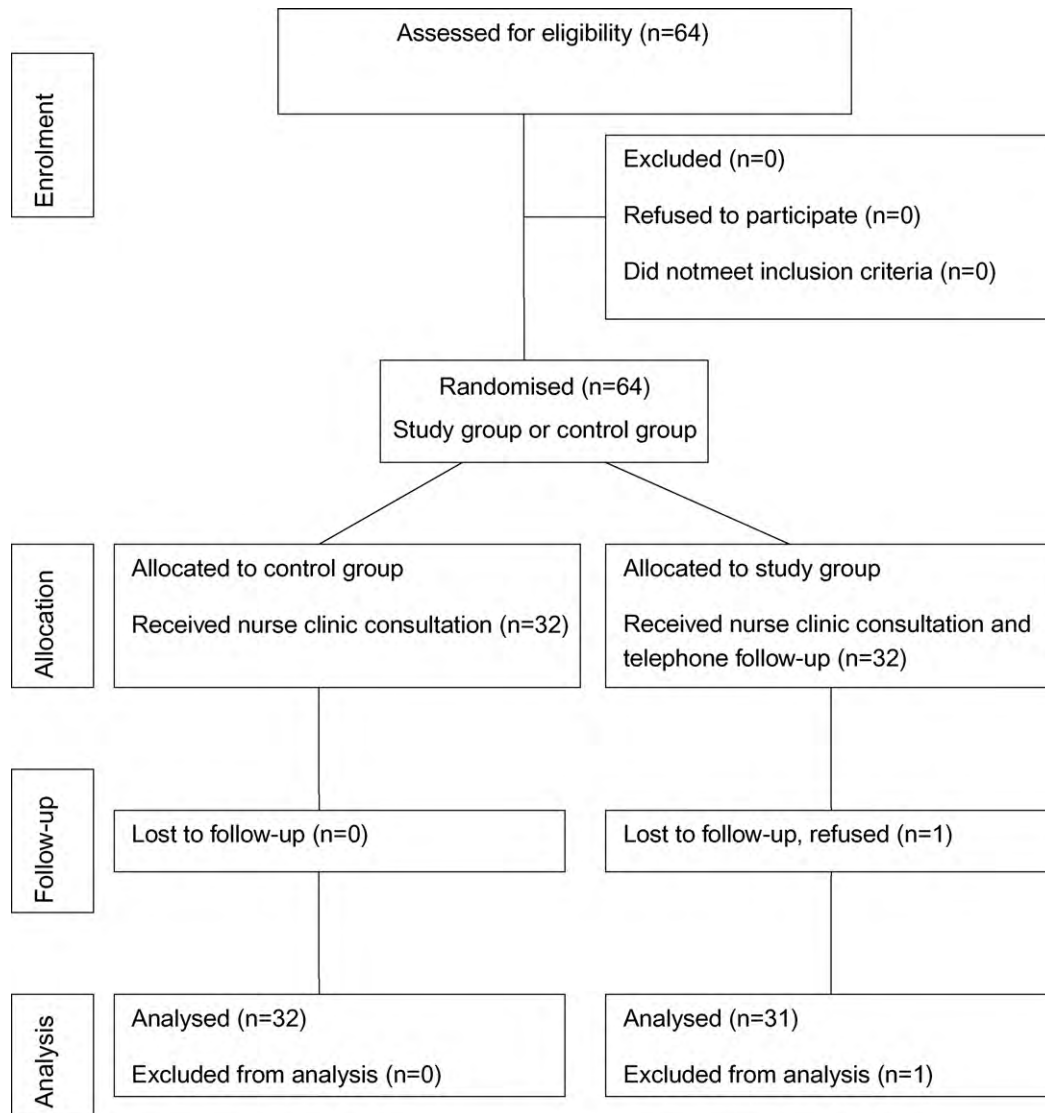
Ethical Approval

By the Hong Kong Polytechnic University and the Hospital Authority, Hong Kong.

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Appendix A. Flow chart of the study



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